习题 15.1

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- (1)
  - (i)

    $f(x)$ 在 $[-\pi,\pi]$ 上是按段光滑的，可以展开为傅里叶级数

$$
\begin{aligned}
a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} xdx \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} xdx \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) cos\, nxdx \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} x cos\, nxdx \\
&= 0
\end{aligned}
$$

注：$xcos\,nx$ 是奇函数。

$$\begin{aligned} b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)sin\,nxdx \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi} xsin\,nxdx \\ &= \frac{1}{\pi}(-\frac{1}{n})\int_{-\pi}^{\pi} xd(cos\,nx) \\ &= -\frac{1}{n\pi}(xcos\,nx\Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} cos\,nx\,dx) \\ &= -\frac{1}{n\pi}[\pi(-1)^n - (-\pi)(-1)^n - 0] \\ &= (-1)^{n+1}\frac{2}{n} \end{aligned}$$

于是

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cosnx + b_n sinnx = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2}{n}sinnx$$

由于 $f(x)$ 在 $(-\pi, \pi)$ 上连续，可知

$$f(x) = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2}{n}sinnx,\ x \in (-\pi + 2k\pi, \pi + 2k\pi)$$

由收敛定理可知，$x = \pm\pi$ 时

$$\begin{aligned} \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2}{n}sinnx &= \frac{f(\pi+0) + f(\pi-0)}{2} \\ &= \frac{\pi + (-\pi)}{2} \\ &= 0 \end{aligned}$$

- (ii) 略

- (2)

  - (1)

$f(x)$ 在 $[-\pi,\pi]$ 上是按段光滑的，可以展开为傅里叶级数，有

$$
\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) cos\, nx dx \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 cos\, nx dx \\
&= \frac{1}{n\pi}\int_{-\pi}^{\pi} x^2 d(sin\, nx) \\
&= \frac{1}{n\pi}\left(x^2 sin\, nx\Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} 2x\, sin\, nx dx\right) \\
&= \frac{1}{n\pi}\left(0 - \int_{-\pi}^{\pi} 2x\, sin\, nx dx\right) \\
&= -\frac{2}{n\pi}\int_{-\pi}^{\pi} x\, sin\, nx dx \\
&= -\frac{2}{n\pi}(-\frac{1}{n})\int_{-\pi}^{\pi} x\, d(cos\, nx) \\
&= \frac{2}{n^2\pi}\left(x\, cos\, nx\Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} cos\, nx dx\right) \\
&= \frac{2}{n^2\pi}\left[(-1)^n 2\pi - 0\right] \\
&= (-1)^n \frac{4}{n^2}
\end{aligned}
$$

$$
\begin{aligned}
a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) dx \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 dx \\
&= \frac{1}{\pi}\left(\frac{1}{3}x^3\Big|_{-\pi}^{\pi}\right) \\
&= \frac{1}{\pi}\frac{2}{3}(\pi)^3 \\
&= \frac{2}{3}(\pi)^2
\end{aligned}
$$

$$
\begin{aligned}
b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) sin\, nx dx \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 sin\, nx dx \\
&= 0
\end{aligned}
$$

故

$$
\begin{aligned}
f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cosnx + b_n sinnx \\
&= \frac{(\pi)^2}{3} \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2} cos\, nx,\ x \in (-\pi, \pi)
\end{aligned}
$$

注： $x = \pm\pi$ 时，

$$
\begin{aligned}
\frac{f(\pi+0)+f(\pi-0)}{2} = \pi^2 &= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2} cos\, n\pi \\
&= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}
\end{aligned}
$$

经过变形，我们有

$$
\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}
$$

- (2)

  $f(x)$ 在 $[0, 2\pi]$ 上是按段光滑的，可以展开为傅里叶级数，有

$$
\begin{aligned}
a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x)dx \\
a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) cos\, nxdx \\
b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) sin\, nxdx
\end{aligned}
$$

  计算略

- (3)

$$
f'(x) = \begin{cases} a, & -\pi < x < 0 \\ b, & 0 < x < \pi \end{cases}
$$

又因为

$$
\lim_{x \to 0^-} f'(x) = a
$$
$$
\lim_{x \to 0^+} f'(x) = b
$$

所以 $f(x)$ 在 $[-\pi,\pi]$ 上是分段光滑的，可以展开为傅里叶级数

$$\begin{aligned}a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx\\ &= \frac{1}{\pi}\left(\int_{-\pi}^{0} ax\,dx + \int_{0}^{\pi} bx\,dx\right)\\ &= \frac{b-a}{2}\pi\end{aligned}$$

$$\begin{aligned}a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)cos\,nxdx\\ &= \frac{1}{\pi}\left(\int_{-\pi}^{0} axcos\,nx\,dx + \int_{0}^{\pi} bxcos\,nx\,dx\right)\\ &= \frac{a-b}{n^2\pi}[1-(-1)^n]\end{aligned}$$

$$\begin{aligned}b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)sin\,nxdx\\ &= \frac{1}{\pi}\left(\int_{-\pi}^{0} axsin\,nx\,dx + \int_{0}^{\pi} bxsin\,nx\,dx\right)\\ &= (-1)^n\frac{a+b}{n}\end{aligned}$$

故

$$\begin{aligned}&\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cosnx + b_n sinnx\\ &= \frac{b-a}{4}\pi + \sum_{n=1}^{\infty}\frac{a-b}{n^2\pi}[1-(-1)^n]cosnx + (-1)^n\frac{a+b}{n}sinnx\end{aligned}$$

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提示：利用第九章 §5 习题 6；